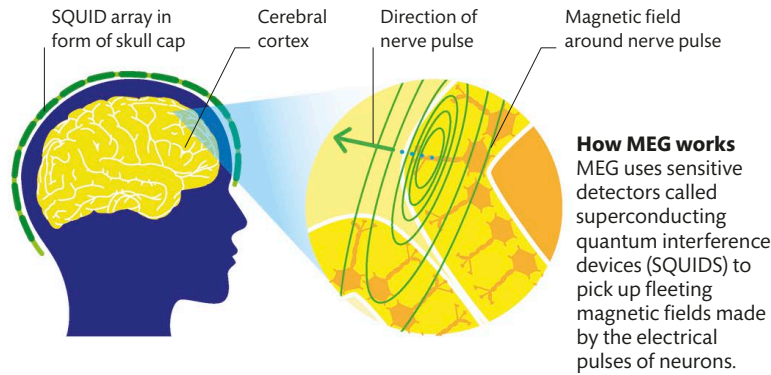




MEG

In addition to making electrical activity, the brain produces a faint magnetic field. This is detected by a magnetoencephalography (MEG) machine and can be used to create a real-time account of activity in the cerebral cortex. MEG is limited by the weakness of the brain's magnetism, but the technique can detect rapid fluctuations in brain activity, which take place over a few thousandths of a second, better than other monitoring systems.

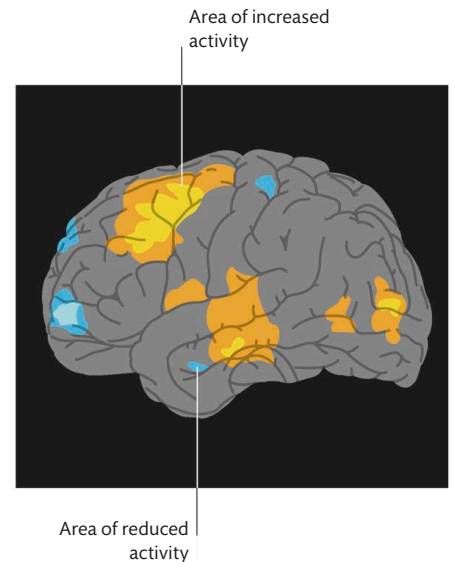
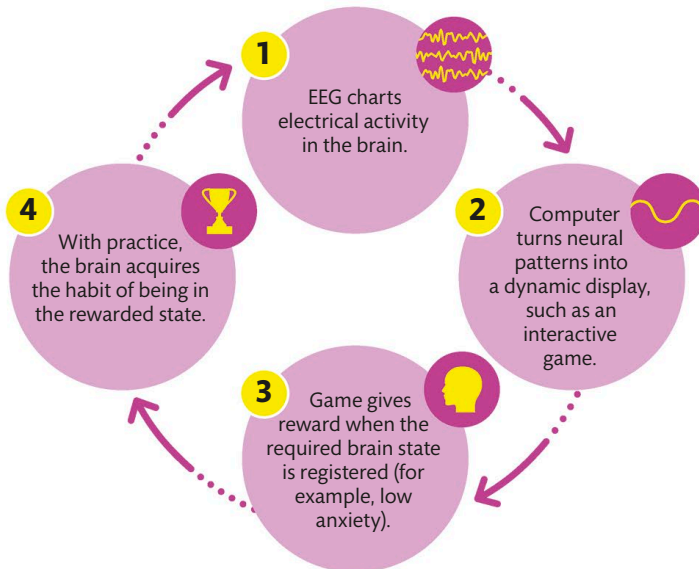


Functional MRI and diffusion tensor imaging

MRI (see pp.40–41) can be extended to collect information about what the brain is doing. Functional MRI (fMRI) scanning tracks the flow of blood through the brain, specifically showing where it is giving oxygen to neurons and thus indicating which regions are active in real time. Subjects are asked to carry out mental and physical tasks while monitored by fMRI to create a functional map of the brain and spinal cord that combines anatomy with activity levels. Diffuse tensor imaging (DTI) also uses MRI but tracks the natural movement of water through brain cells. It is used to build up a map of the white-matter connections within the brain.

NEUROFEEDBACK

This form of cognitive therapy uses an EEG to create a feedback loop between a person's mental state and their brain activity. This makes it easier for people to learn ways to control unwanted mental activity, such as anxiety.



Interpreting an fMRI image

An fMRI scan begins with establishing a baseline of activity in the brain. The scan then shows up regions that fluctuate from this baseline, allowing researchers to figure out which areas are excited or inhibited during particular tasks.